Lebanese University - Faculty of Sciences III
Computer science department
I2207 – Computer Architecture

Assignment
Cryptographic Memory Maze

Implement a MIPS32 program that manipulates memory-mapped data structures using bitwise operations and student-specific cryptographic patterns.

Deadline: 31st May 2025

Statement:

Design a memory system that:

1. Accepts an 8-digit student ID number as input (e.g., yyyymmdd)
2. Generates a unique 16×16 memory maze where:
   - Walls are determined by (ID % prime_number) for sequential primes < 50
   - Data paths contain 32-bit values encrypted using your ID
3. Implements a stateful decoder that:
   - Uses bit rotation patterns derived from your ID
   - Requires 4 separate memory buffers (A/B/C/D) with XOR-chain dependencies

Guidance:

Wall generation	ID % primes == 0 then "the cell is a wall".
Datapath encryption	position_value = (position_x << 16) OR position_y) temp = position_value XOR ID cell_value = ROTATE temp by (ID % 32)